

Resource_Module_model_summary.txt

TBD2 pipeline output - rendered to PDF

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.38030394934865536

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9582314936921804

Scenario_9: 0.8038222834527426

Scenario_8: 0.6386992184153006

Scenario_7: 0.4897011323447945

Scenario_6: 0.3610822487815188

Scenario_5: 0.2517311237794721

Scenario_4: 0.15915817873087432

Scenario_3: 0.0901217409995456

Scenario_2: 0.03943721790523324

Scenario_1: 0.011054855384891085

Average MSE across models is 9.385521644319177e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: RSU10119, RST10117, RSL10111. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: RSU10108, RCM10111, RSU10107. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: RDA10107, RSU10103, RCD10113. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- RSU10119: Root Cause Analysis (RCA) Completion Rate
- RST10117: Resource Allocation Accuracy
- RSL10111: Resource Activation Success Rate

Scenario_9 (Avg Risk=0.804):

- RSU10108: Escalation Rate
- RCM10111: Resource Cost Efficiency
- RSU10107: Mean Time to Repair (MTTR)

Scenario_8 (Avg Risk=0.639):

- RDA10107: Data Security Incident Rate
- RSU10103: Incident Resolution Time
- RCD10113: Resource Availability Post-Deployment

Baseline Predictor (Mean of RSRC_001):

MSE: 1.6345371898222225

R-squared: 0.0

RMSE: 1.278490199345393

MAPE: 1.7032619865151577%

PyTorch Model Evaluation:

MSE: 0.0005518431050731974

R-squared: 0.9996623857147398

RMSE: 0.02349134106587356

MAPE: 0.030786957166234717%

Model Evaluation Metrics:

MSE: 0.0005518431050731974

RMSE: 0.02349134106587356

R²: 0.9996623857147398

MAPE: 0.030786957166234717%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of Resource Module Risk Analysis Summary-----